Volatility Arbitrage via the Variance Risk Premium

Trading the IV–Forecast RV Spread on SPX Options

Group 4
MS&E 244: Statistical Arbitrage

Stanford University

Winter 2026

Motivation & Strategy

Motivation

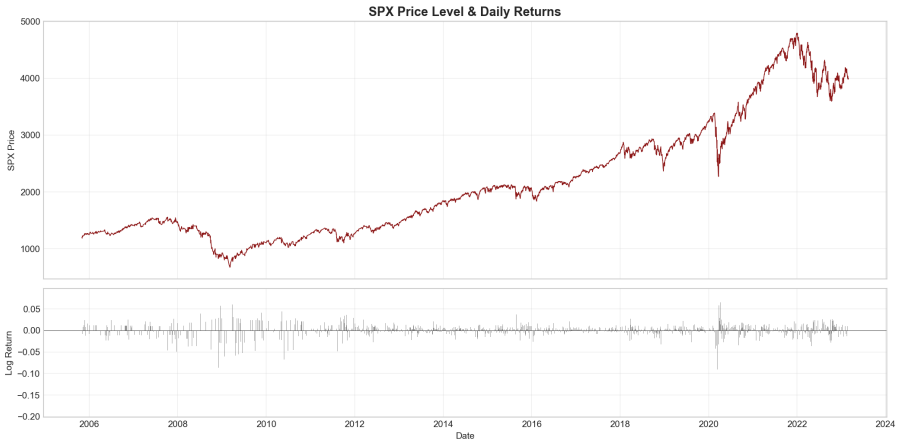
- **VRP:** Implied variance > realized variance.
- Compensates sellers for crash risk.
- **Question:** Is IV–forecast RV spread tradeable?

Strategy

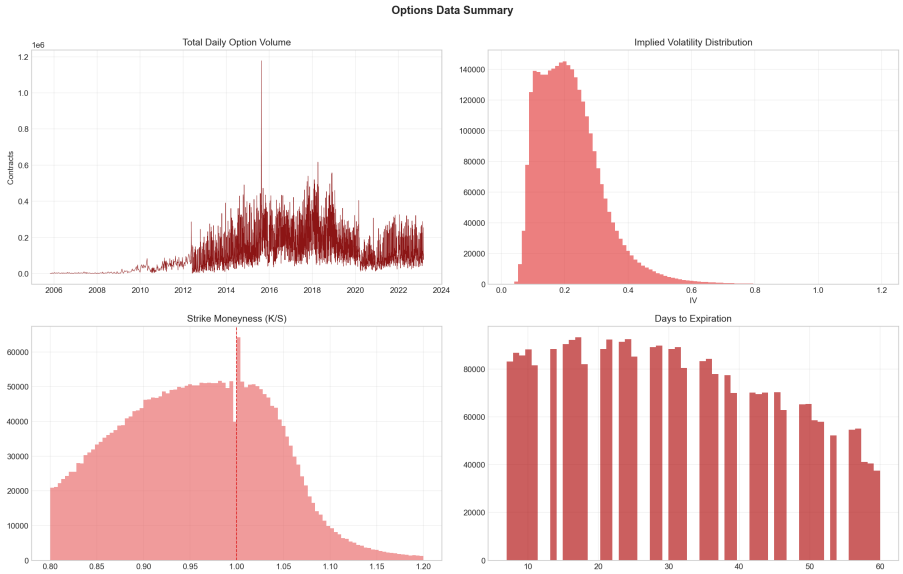
- Delta-hedged ATM straddles on SPX.
- Enter when VRP z-score > 1 (short vol) or < −1 (long vol).
- Hold to expiry (1–5d for weeklies); 25% stop per trade (early exit & expiry); $1M

VRP z-score and discrete signal.

Data: SPX & Options



SPX price and daily log returns.

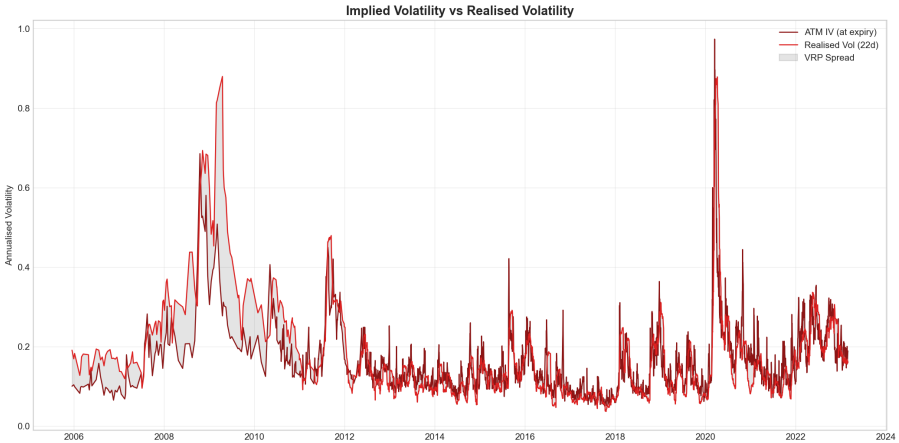


Options: volume, IV, moneyness, DTE. OptionMetrics 2005–2023; filters on bid, spread, DTE 7–60, moneyness $\pm 20\%$.

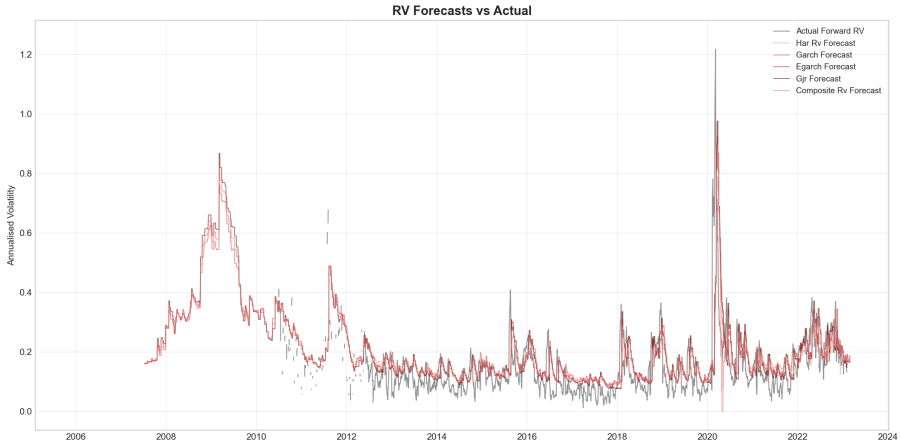
Strategies Compared (Methods & Models)

1. **VRP (Variance Risk Premium)** — Level signal
   - IV: ATM at expiry (1–5d); RV: HAR/GARCH/GJR composite forecast
   - Signal: $z = (\text{IV} - \text{RV})/\sigma$; enter $|z| > 1$; short vol when IV rich
2. **Var Swap** — Model-free IV
   - IV: MFIV at expiry (variance swap rate from full option surface)
   - Same RV forecast and z-score rules; captures entire implied variance

IV vs RV & RV Forecasts

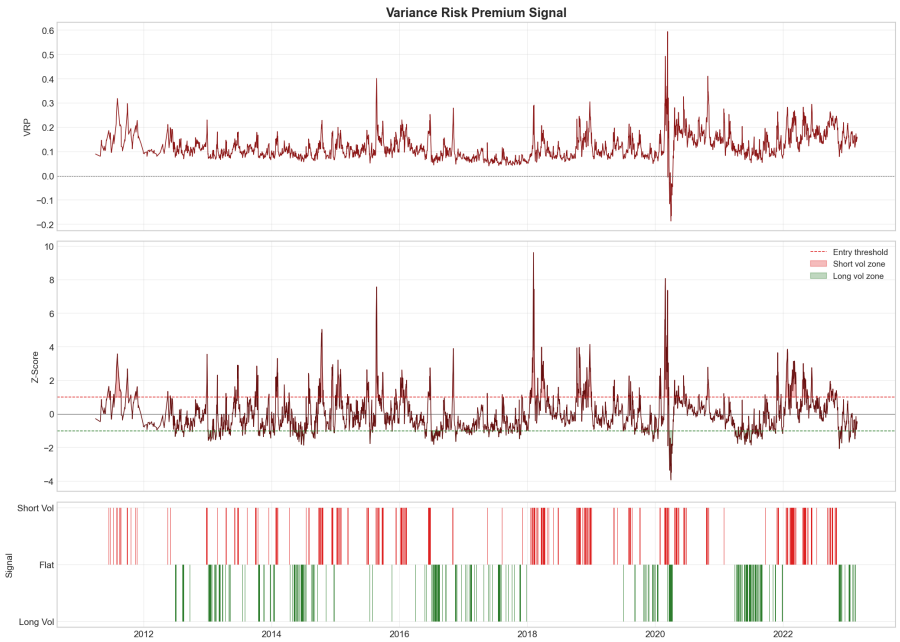


ATM IV (at expiry) vs realized vol; spread = VRP raw input.



HAR-RV, GARCH, GJR vs actual forward RV; composite = equal-weight.

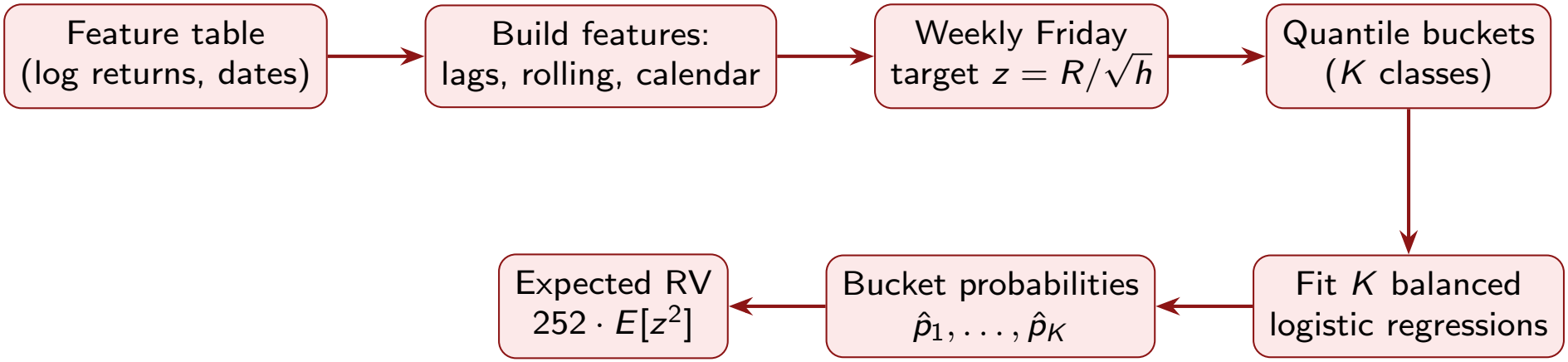
Signals: VRP & Surface



VRP level, z-score, and ± 1 entry bands.

Skew and term-structure signals.

Logistic Regression: Methodology



Workflow: predict distribution of weekly (Friday-expiry) returns; convert to expected realised variance and use as RV forecast in IV–RV signal.

Target: cumulative log return to Friday, scaled by $\sqrt{\text{days_to_Friday}}$ for comparable horizon. One classifier per quantile (balanced class weights); probabilities normalized to sum to 1; $E[z^2] = \sum_k \hat{p}_k \cdot (\text{median}_k)^2$.

Logistic Regression: Models & Features

Features (inputs)

- **Lagged returns:** log-return at lags 1, 2, 3, 5, 10, 22; absolute lags (volatility proxy).
- **Rolling:** 5d and 22d rolling mean, std, and mean of $|r|$ (lagged by 1 to avoid look-ahead).
- **Calendar:** weekday, days_to_Friday (horizon for weekly expiry).

Models

- **Target:** Forward return to Friday expiry, standardized $z = R_{\text{cum}}/\sqrt{h}$; bucketed into K quantiles (e.g. $K = 4$ or 5).
- **One-vs-rest:** K binary logistic regressions (one per bucket), `class_weight='balanced'`, StandardScaler + LBFGS.
- **Output:** Predicted probability per bucket $\rightarrow E[z^2] \rightarrow$ annualized expected realised variance/volatility; used as RV forecast in same IV–RV (VRP) signal and backtest.

Out-of-sample: expanding window (e.g. 252-day train), refit each period; same entry/exit rules as main VRP strategy.

Logistic Regression: Results vs VRP

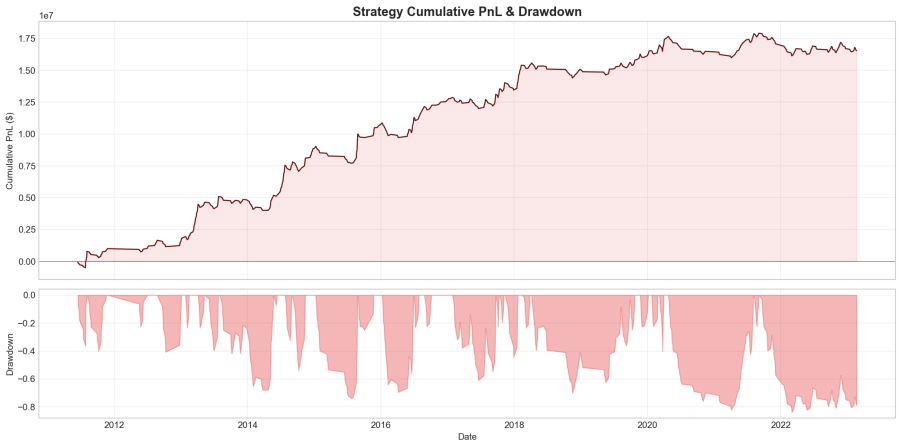
Quantile sweep (same IV–RV signal, backtest; 25% stop per trade)

Strategy	Trades	Win Rate	Sharpe	Total PnL	Max DD
VRP (composite RV)	188	46.3%	2.16	\$16.54M	−83.9%
Logistic K=8 (best)	186	47.3%	2.11	\$16.62M	−84.9%
Logistic K=4	165	43.6%	2.02	\$15.09M	−86.2%
Logistic K=3	160	43.8%	1.74	\$12.47M	−88.1%

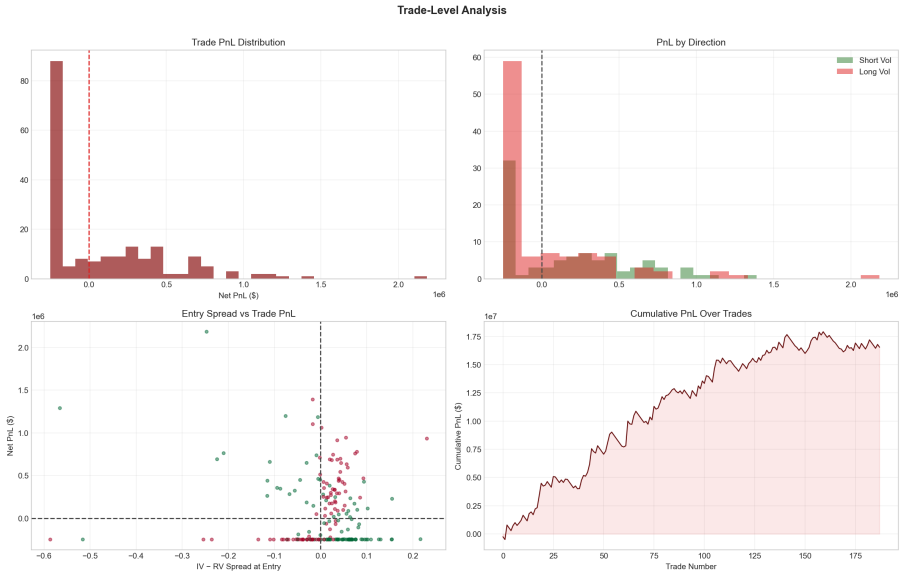
Conclusion:

Composite RV (HAR/GARCH) slightly outperforms logistic; both profitable. Best logistic at K=8.

Cumulative PnL & Trade Analysis

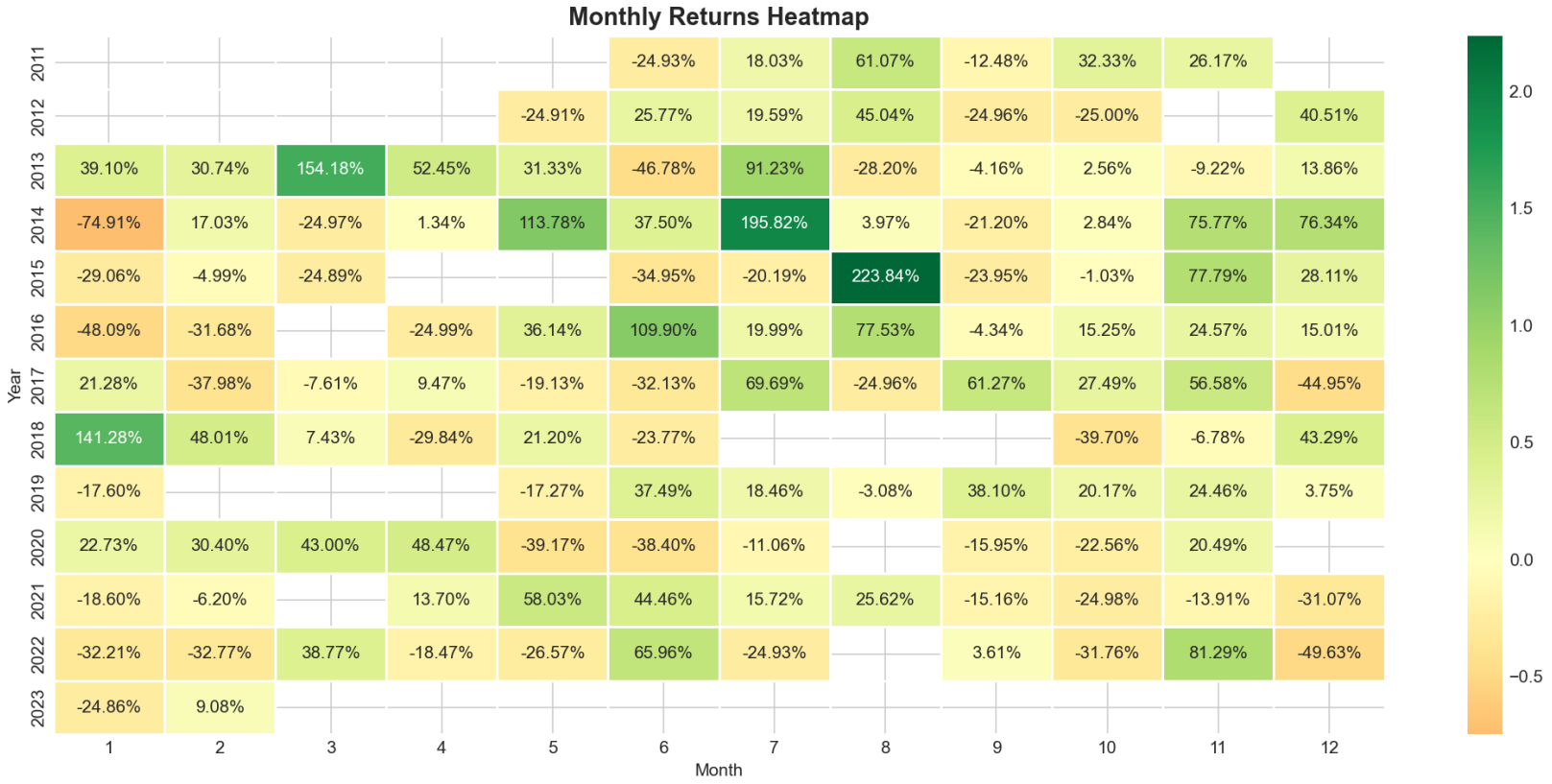


Cumulative PnL and drawdown (25% stop per trade, early + expiry).



PnL distribution, IV vs RV at entry, direction breakdown.

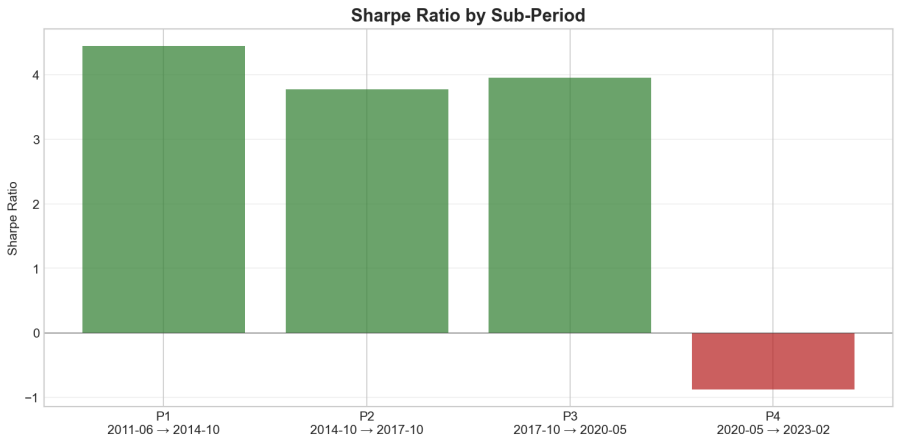
Monthly Returns & Key Metrics



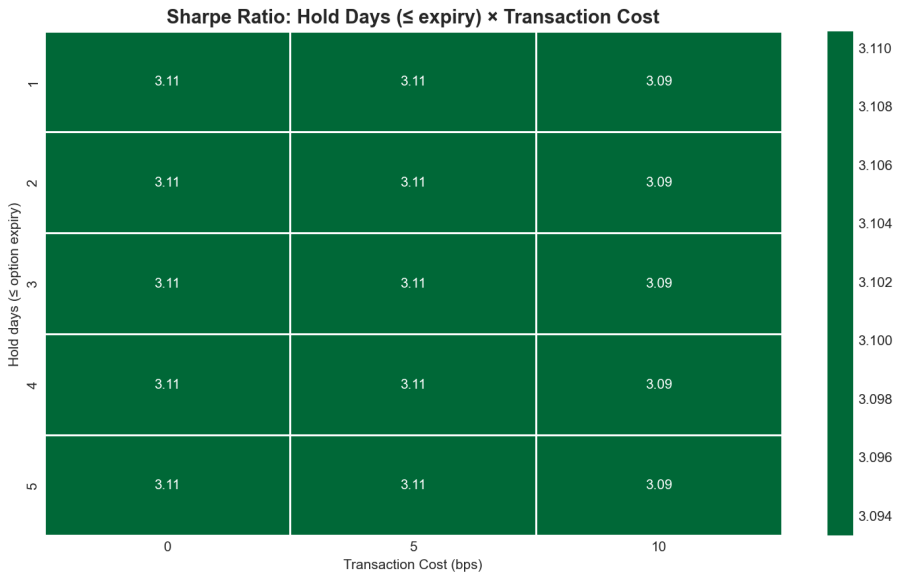
Monthly return heatmap.

Full-sample (VRP, 25% stop): 188 trades, 46.3% win, \$16.54M total PnL; Sharpe 2.16, Sortino 4.43, Calmar 2.91; Ann. return 244%, vol 67.1%, max DD −83.9%; Benchmark (SPX): Sharpe 0.66; alpha 324%, info ratio 3.41.

Robustness & Parameter Sensitivity (Pipeline)

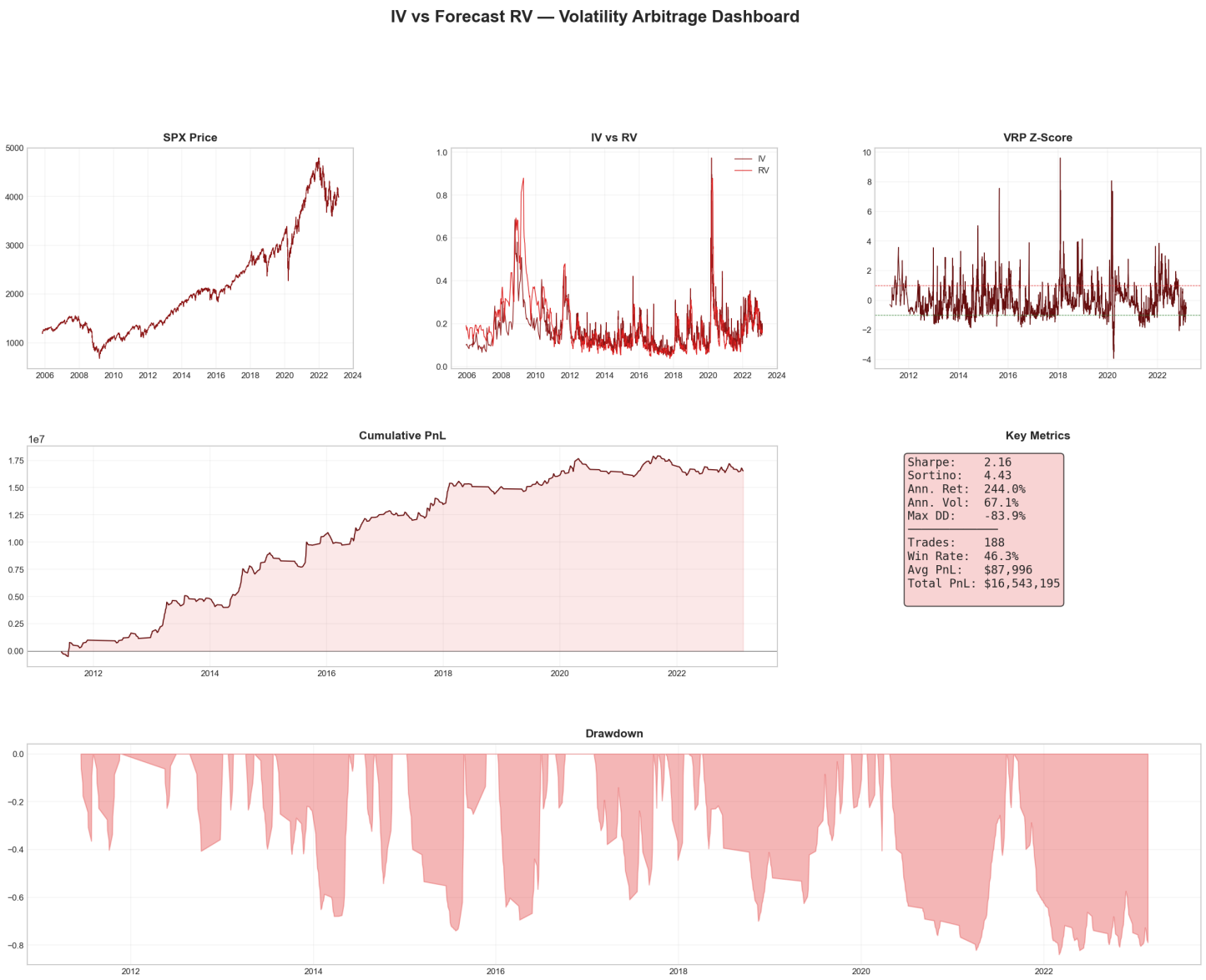


Sharpe by sub-period.



Sharpe across hold days × cost (bps).

Summary Dashboard



Six-panel overview: SPX, IV vs RV, VRP signal, cumulative PnL, metrics, drawdown.

Strategy Comparison

Strategy	Trades	Win%	Total PnL	Sharpe	PSR(SR>0)	Alpha	Max DD
VRP (ATM IV	188	46%	$16.54M	2.16	100%	324.0%	-83.9%
Var Swap (MFI	190	48%	$24.89M	2.54	100%	635.9%	-82.3%

VRP vs Var Swap: PnL, Sharpe, drawdowns.

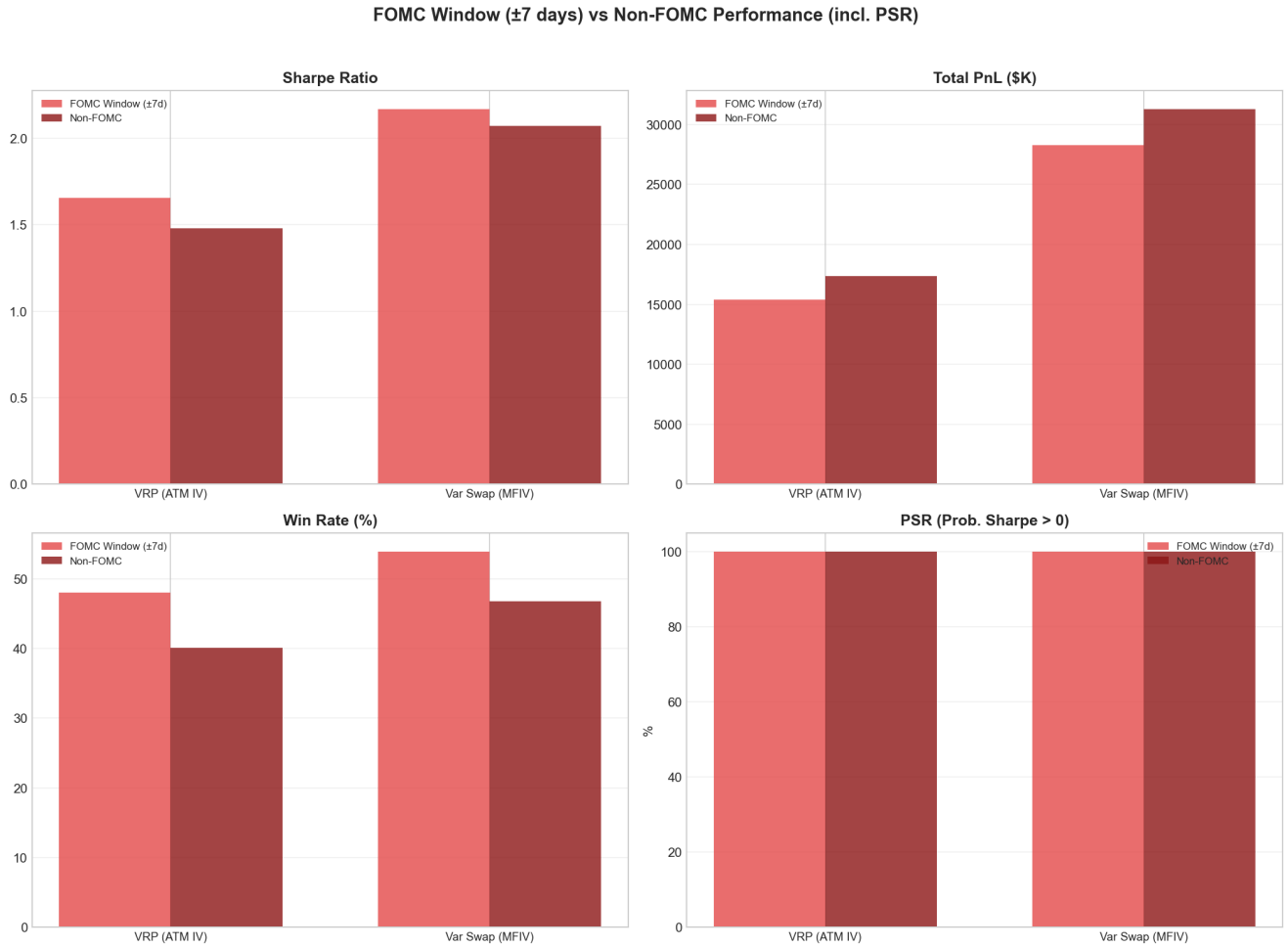
Var Swap leads: Var Swap: 190 trades, Sharpe 2.54, $24.89M, max DD −82.3%; VRP: 188 trades, Sharpe 2.16, $16.54M, max DD −83.9%. 25% stop per trade caps losses.

Results Comparison: All Strategies

Strategy	Trades	Win Rate	Sharpe	Total PnL	Max DD
VRP (ATM IV, composite RV)	188	46.3%	2.16	\$16.54M	−83.9%
Var Swap (MFIV)	190	48.4%	2.54	\$24.89M	−82.3%
Logistic (RV forecast, K=8)	186	47.3%	2.11	\$16.62M	−84.9%

Same backtest: delta-hedged straddles, \$1M notional, hold to expiry (1–5d), 25% stop per trade (early + expiry), 5 bps cost.

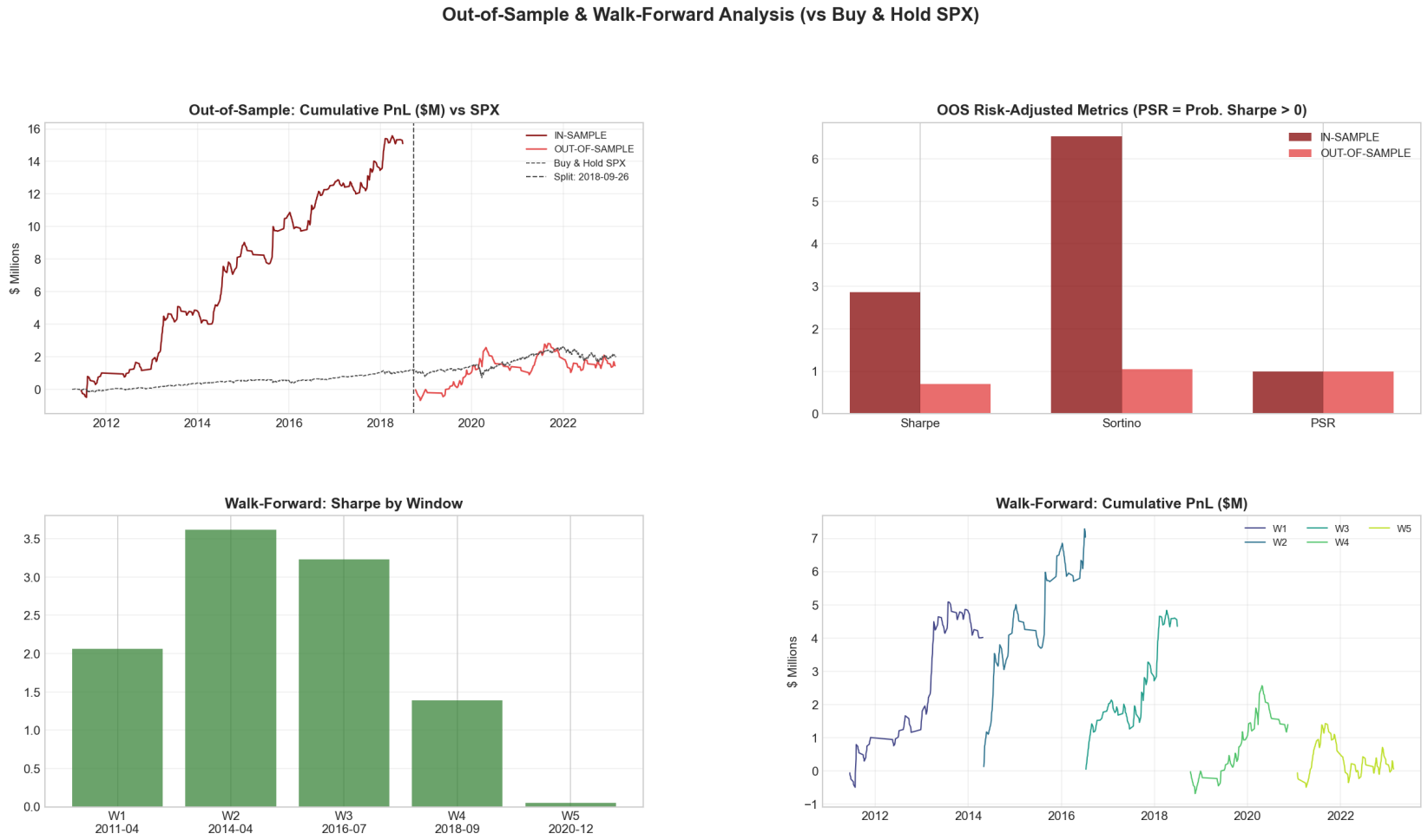
FOMC Window Analysis



FOMC ±7d vs non-FOMC across strategies.

FOMC vs non-FOMC (VRP): FOMC: 79 trades, 48% win, $15.37M, Sharpe 1.65; Non-FOMC: 142 trades, 40% win, $17.35M, Sharpe 1.48. Both profitable.

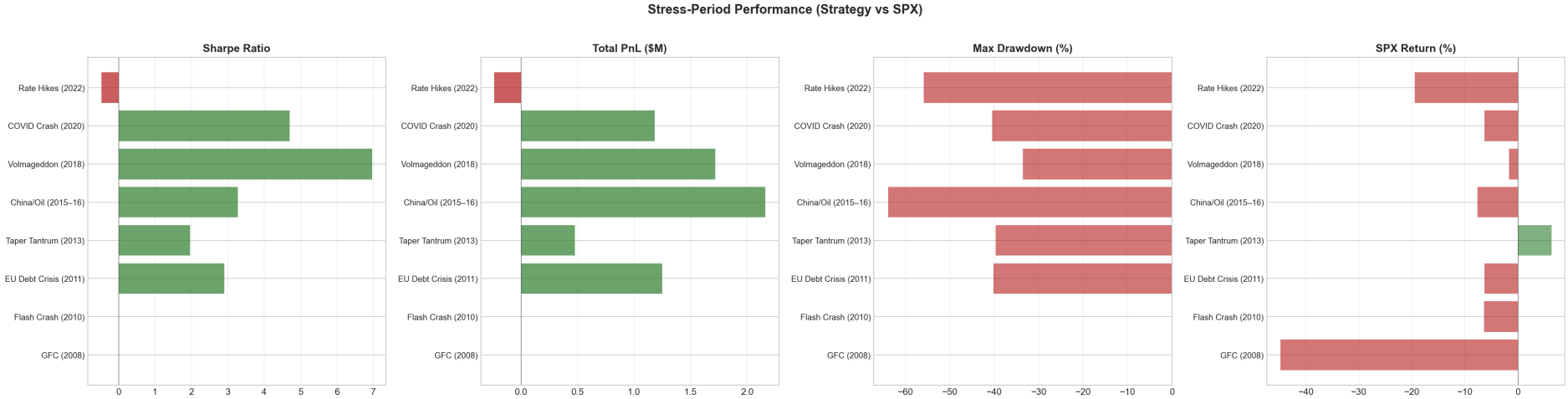
Out-of-Sample & Walk-Forward



OOS cumulative PnL; walk-forward Sharpe by window.

OOS (60/40 split): In-sample: 115 trades, Sharpe 2.86, $15.10M; Out-of-sample: 73 trades, Sharpe 0.69, $1.44M. **Walk-forward:** 5 of 5 windows profitable.

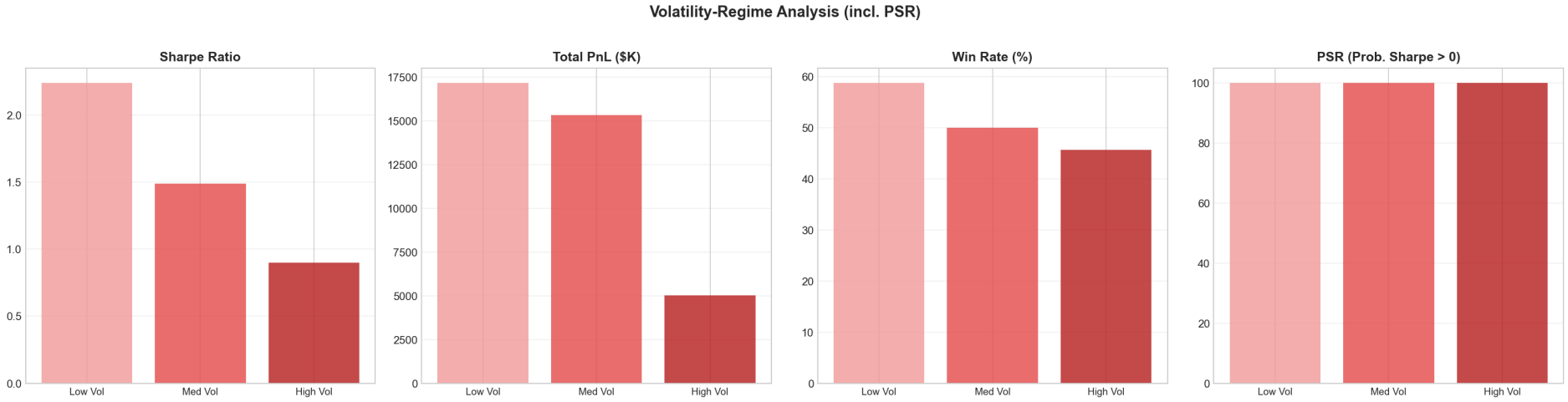
Stress Periods



Sharpe, PnL, max DD by stress episode.

Selected episodes: Volmageddon, China/Oil, COVID: strong (Sharpe 6.96, 3.27, 4.70); 2022 rate hikes: negative (Sharpe −1.07).

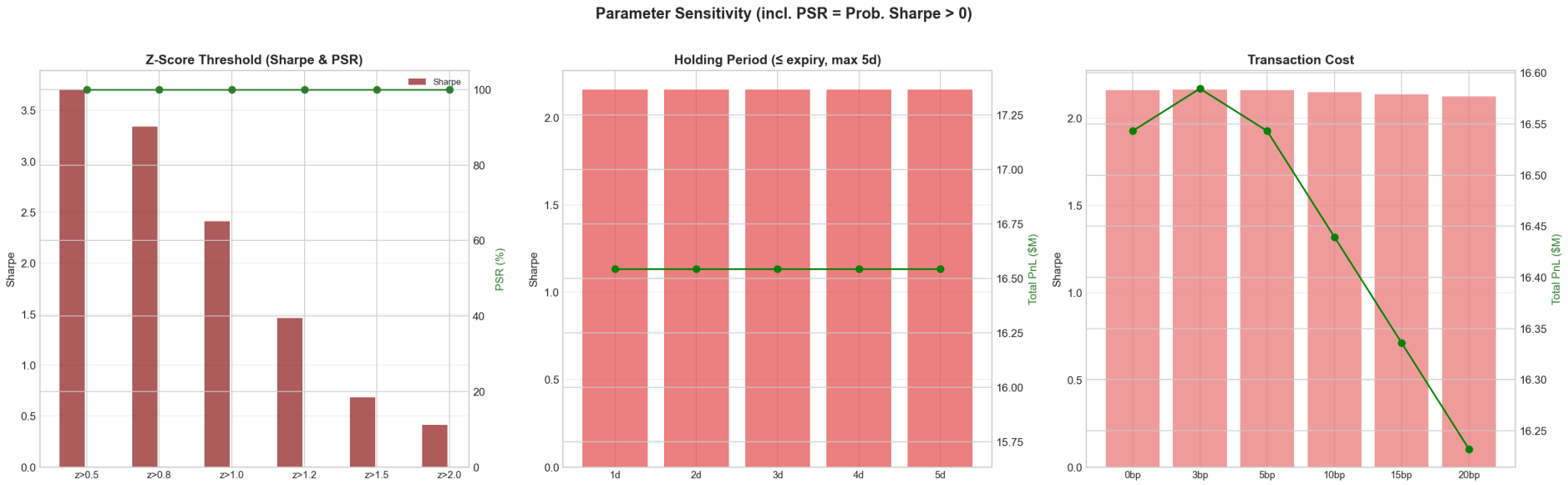
Volatility Regime



Performance by trailing 252d RV tercile.

Regime (VRP): Low vol: \$17.17M, Sharpe 2.24; Medium vol: \$15.32M, Sharpe 1.49; High vol: \$5.02M, Sharpe 0.90. Profitable across regimes; best in low vol.

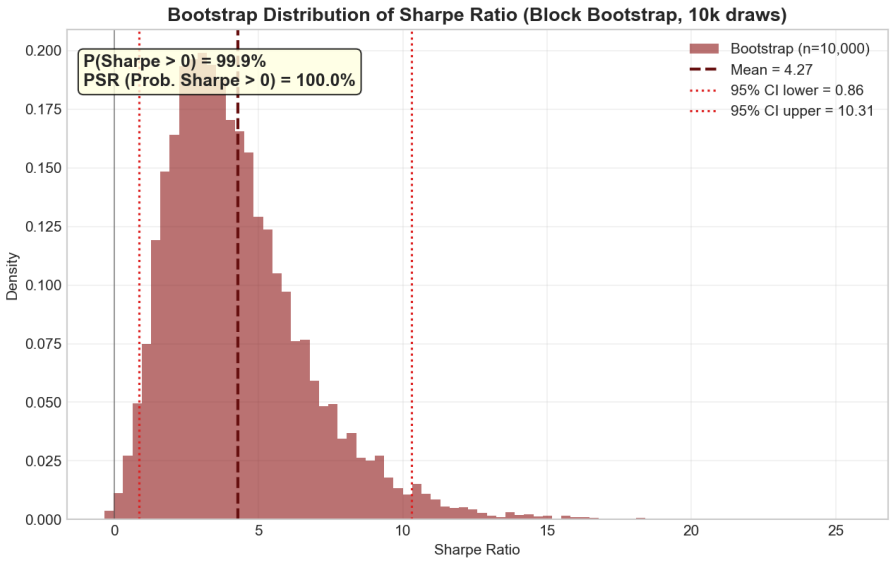
Parameter Sensitivity (Robustness)



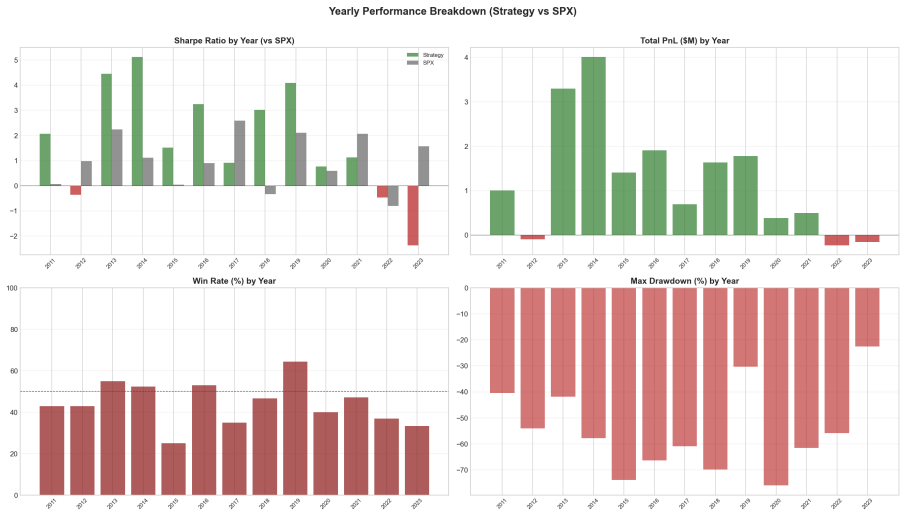
Z-threshold, hold days, transaction cost.

Findings: Z-threshold: Sharpe positive across grid; degrades at $z > 1.5$. Hold: stable at 2.16 for 1–5d (weeklies). Cost: robust 0–20 bps.

Bootstrap Sharpe & Yearly Breakdown



Block bootstrap (10k, 22-day blocks); 95% CI, P(Sharpe > 0).



Annual Sharpe, PnL, win rate, max DD (2011–2023). Profitable 11 of 13 years; \$16.54M cumulative.

Summary

- VRP (IV − forecast RV) is **tradeable** on SPX with delta-hedged straddles.
- Sharpe 2.16, \$16.54M PnL (188 trades); Var Swap: Sharpe 2.54, \$24.89M (190 trades). Max DD $\approx$ 84% (25% stop per trade caps losses).
- Var Swap outperforms VRP on Sharpe and PnL; strong in stress (Volmageddon, COVID); regime-dependent (low-vol best).
- **Logistic regression:** alternative RV forecast (quantiles $\rightarrow E[z^2] \rightarrow$ expected vol); Sharpe 2.11, \$16.62M at K=8; composite RV slightly better.
- Robust to threshold and costs; hold to expiry (1–5d for weeklies).

Limitations & Downfalls

- **Tail risk:** ≈84% max DD; 25% stop per trade (early + expiry) caps each loss; assumes executable at cap.
- **Regime-dependent:** Best in low vol (Sharpe 2.24); weaker in high vol (Sharpe 0.90).
- **FOMC:** Both FOMC and non-FOMC windows profitable (Sharpe 1.65 vs 1.48).
- **Statistical uncertainty:** Bootstrap P(Sharpe > 0) $= 99.9\%$; 95% CI [0.86, 10.31].
- **Single underlying:** SPX only; one position at a time; limited capacity.
- **Simplified hedging:** Daily delta hedge; no intraday gamma; costs may be optimistic in stress.

Thank you

Questions?